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Shijun ZHANG

Assistant Professor

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Appointments

Assistant Professor (Tenure-Track), Jul 2024 ∼ Present
Hong Kong Polytechnic University (PolyU), Hong Kong SAR, China

Phillip Griffiths Assistant Research Professor, Aug 2022 ∼ Jun 2024
Duke University, United States
Mentors: **Jianfeng Lu** and **Hongkai Zhao**

Research Fellow, National University of Singapore, Singapore Jan 2021 ∼ Jul 2022
Mentor: **Zuowei Shen**

Education

Ph.D. in Mathematics, National University of Singapore, Singapore Aug 2016 ∼ Jan 2021
Thesis: *Deep neural network approximation via function compositions* [PDF, URL]
Supervisors: **Zuowei Shen** and **Haizhao Yang**

B.S. in Mathematics, Wuhan University, China Sep 2012 ∼ Jun 2016
Thesis supervisor: **Xiliang Lv**

Awards

The 2026 Frontiers of Science Award (Mathematics), International Congress for Basic Science (ICBS 2026), Award Medal, URL.

Departmental Best Paper Award, AMA, PolyU, 2024-2025, Award Trophy.

The EASIAM (East Asia section of SIAM) Student Paper Prize (First Prize), 2020-2021, URL.

Teaching

Will teach **Learning Theory for AI** (AMA 579) in 2026/27 Sem2.

Operations Research Methods (AMA 3820), PolyU 2025/26 & 2026/27 Sem1 & Sem2
Instructor, **Syllabus**

Optimization Methods (AMA 4850), PolyU 2024/25 & 2025/26 Sem2
Instructor, **Syllabus**

Mathematical Numerical Analysis (Math 361S), Duke University Spring 2024
Instructor, **Syllabus**

Matrices and Vectors (Math 218D-2), Duke University Fall 2023
Teaching assistant

Publications

[number] Author(s). *Paper title.* Journal or conference reference. [Links]

* Corresponding author † Equal contribution

Preprint(s)

[22] **Shijun Zhang**. *Sharp approximation rates for neural networks with affine latent parameterizations.* Submitted. [arXiv]

[21] Baicheng Li, Haizhao Yang, **Shijun Zhang**. *Sobolev approximation by fixed-size neural networks with arbitrary accuracy.* Submitted. [arXiv]

[20] **Shijun Zhang***, Zuowei Shen, Yuesheng Xu. *Layer-wise geometric approximation rates for deep networks.* Submitted. [arXiv]

[19] Zeyu Li, **Shijun Zhang**, Tieyong Zeng, Feng-Lei Fan. *Neural network approximation: a view from polytope decomposition.* Submitted. [arXiv]

[18] **Shijun Zhang***, Zuowei Shen, Yuesheng Xu. *Multigrade neural network approximation.* Submitted. [arXiv]

[17] Hanyu Pei, Jing-Xiao Liao, Qibin Zhao, Ting Gao, **Shijun Zhang**, Xiaoge Zhang, Feng-Lei Fan. *Neuronseek: on stability and expressivity of task-driven neurons* Submitted. [arXiv]

Published (Accepted)

[16] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Fourier multi-component and multi-layer neural networks: unlocking high-frequency potential.* Neural Networks, 204:109268, 2026. [arXiv, Journal]

[15] Fenglei Fan, Juntong Fan, Dayang Wang, Jingbo Zhang, Zelin Dong, **Shijun Zhang**, Ge Wang, Tieyong Zeng. *Hyper-compression: model compression via hyperfunction.* IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 48(4):8813–8830, 2026. [arXiv, Journal]

[14] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Structured and balanced multi-component and multi-layer neural networks.* **SIAM Journal on Scientific Computing**, 47(5):C1059-C1090, 2025. [arXiv, Journal]

[13] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Why shallow networks struggle to approximate and learn high frequencies.* **Information and Inference: A journal of the IMA**, 14(3):iaaf022, September 2025. [arXiv, Journal]

[12] Qianchao Wang†, **Shijun Zhang**†, Dong Zeng, Zhaoheng Xie, Hengtao Guo, Feng-Lei Fan, Tieyong Zeng. *Don't fear peculiar activation functions: euaf and beyond.* **Neural Networks**, 186:107258, June 2025. [arXiv, Journal]

[11] **Shijun Zhang***, Jianfeng Lu, Hongkai Zhao. *Deep network approximation: beyond ReLU to diverse activation functions.* **Journal of Machine Learning Research**, 25(35):1–39, 2024. [arXiv, Journal]

[10] **Shijun Zhang***, Jianfeng Lu, Hongkai Zhao. *On enhancing expressive power via compositions of single fixed-size ReLU network.* Proceedings of the 40th International Conference on Machine Learning (**ICML 2023**), PMLR 202:41452–41487, 2023. [arXiv, Poster, Conference]

[9] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Neural network architecture beyond width and depth.* Advances in Neural Information Processing Systems (**NeurIPS 2022**), 35:5669–5681, 2022. [arXiv, Poster, Conference]

[8] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Deep network approximation in terms of intrinsic parameters.* Proceedings of the 39th International Conference on Machine Learning (**ICML 2022**), PMLR 162:19909–19934, 2022. [arXiv, Spotlight, Conference]

[7] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Deep network approximation: achieving arbitrary accuracy with fixed number of neurons.* **Journal of Machine Learning Research**, 23(276):1–60, September 2022. [arXiv, Journal]

[6] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Optimal approximation rate of ReLU networks in terms of width and depth.* **Journal de Mathématiques Pures et Appliquées**, 157:101–135, January 2022. [arXiv, Journal]

[5] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *neural network approximation: three hidden layers are enough.* **Neural Networks**, 141:160–173, September 2021. [arXiv, Journal]

[4] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network with approximation error being reciprocal of width to power of square root of depth.* **Neural Computation**, 33(4):1005–1036, April 2021. [arXiv, Journal]

[3] Jianfeng Lu, Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network approximation for smooth functions.* **SIAM Journal on Mathematical Analysis**, 53(5):5465–5506, September 2021. [arXiv, Journal]

[2] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network approximation characterized by number of nuerons.* **Communications in Computational Physics**, 28(5):1768–1811, November 2020. [arXiv, Journal]

[1] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Nonlinear approximation via compositions.* **Neural Networks**, 119:74–84, November 2019. [arXiv, Journal]